

CRTLPyme - Comprehensive Code Analysis

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Executive Summary

CRTLPyme is a **SaaS Point of Sale (POS) system** designed for Chilean small and medium businesses (PYMES). It's built with Next.js 15, PostgreSQL, Prisma ORM, and deployed on Google Cloud Platform (Cloud Run). The system features:

- **Multi-tenant architecture** with subscription-based access
- **Role-based access control** (5 roles: PROVEEDOR, ADMIN, CAJA, INVENTARIO, SOPORTE)
- **Master Product Catalog** for shared products across tenants
- **Complete POS functionality** with cash sessions and inventory management
- **Transbank payment integration** for subscription payments
- **SaaS Admin Dashboard** for platform management
- **CI/CD pipeline** via GitHub Actions

Database Schema Overview (Prisma)

Core Multi-Tenant Tables

1. Tenant (Main tenant entity)

- **Purpose:** Represents each business client
- **Key Fields:**
 - `businessName` , `rut` , `email` , `phone` , `address`
 - `planType` : BASIC, PRO, ENTERPRISE
 - `maxCashiers` , `extraCashiers`
 - `accountStatus` : ACTIVE, TRIAL, SUSPENDED, BLOCKED, CANCELLED
 - `trialStartedAt` , `trialEndsAt`
 - `onboardingCompleted` , `totalRevenue` , `lifetimeMonths`

2. User (Multi-tenant users)

- **Purpose:** All users in the system
- **Key Fields:**
 - `email` , `password` , `firstName` , `lastName`
 - `role` : PROVEEDOR, ADMIN, CAJA, INVENTARIO, SOPORTE
 - `isActive` , `tenantId`
- **Relations:** Links to tenant, sales, cash sessions

3. MasterProduct (Shared product catalog)

- **Purpose:** Global product pool with Chilean products
- **Key Fields:**
 - sku , barcode (EAN-13)
 - name , description , category , brand
 - suggestedPrice , unit , imageUrl
- **Design:** Single source of truth for product data

4. TenantInventory (Per-tenant product inventory)

- **Purpose:** Each tenant's specific product inventory
- **Key Fields:**
 - tenantId , masterProductId
 - customSku , costPrice , salePrice
 - stock , minStock , location
 - customNotes , isActive
- **Relations:** Links to MasterProduct and Tenant

5. Product (DEPRECATED - Legacy)

- **Note:** Marked as products_legacy , kept for compatibility

Sales & Operations Tables

6. CashSession

- Cash register opening/closing with amounts and discrepancies
- Links to user and tenant
- Status: OPEN, CLOSED

7. Sale

- Complete sale records with payment method tracking
- Fields: saleNumber , subtotal , tax , total
- Payment methods: CASH, DEBIT, CREDIT, TRANSFER
- Status: PENDING, COMPLETED, CANCELLED

8. SaleItem

- Individual line items per sale
- Tracks unitPrice , unitCost , quantity , subtotal
- Links to TenantInventory (not Product)

9. StockAdjustment

- Manual inventory adjustments
- Types: PURCHASE, LOSS, CORRECTION, RETURN
- Tracks user who made adjustment

10. InventoryMovement

- Automatic inventory tracking
- Types: ENTRY, EXIT, ADJUSTMENT
- Records all stock changes with reasons

11. FixedExpense

- Business expenses (rent, utilities, etc.)

- Frequency: DAILY, WEEKLY, MONTHLY, YEARLY

SaaS Platform Administration

12. PlatformAdmin

- Platform administrators (not tenant users)
- Roles: SUPER_ADMIN, SUPPORT, BILLING_ADMIN

13. PlatformAdminSession

- Admin session management with tokens

14. TenantManagement

- Tracks tenant account status and actions
- Fields: suspensionReason , blockedReason , riskLevel

15. TenantActionLog

- Audit log of admin actions on tenants

16. SubscriptionPlan

- Available subscription plans
- Fields: name , price , billingCycle , trialDays
- Features: maxUsers , maxProducts , maxSales

17. Subscription

- Active tenant subscriptions
- Status: ACTIVE, TRIAL, CANCELLED, EXPIRED, SUSPENDED
- Tracks billing cycles and renewals

18. SubscriptionPayment

- Payment records with Transbank integration
- Fields: transbankOrderId , transbankToken , transbankBuyOrder
- Status: PENDING, APPROVED, REJECTED, FAILED, REFUNDED

19. PaymentWebhook

- Webhook data from payment provider

20. Refund

- Payment refund tracking

Metrics & Reporting

21-24. Platform Metrics Tables

- PlatformMetrics : Daily platform statistics
- RevenueReport : Revenue analysis by period
- SubscriptionMetrics : Plan-specific metrics
- DashboardSnapshot : Cached dashboard data

Notifications & Email

25-29. Email & Notification Tables

- EmailTemplate : Reusable email templates
- EmailQueue : Pending emails with retry logic
- EmailLog : Email delivery tracking
- NotificationPreference : User notification settings

- `NotificationHistory` : Sent notification records

Audit & Security

30. AuditLog

- Complete audit trail of system changes
- Tracks: action, entity, entityId, oldValues, newValues

Project Structure Analysis

Frontend Routes (app/)

Public Routes

- `/` - Landing page with pricing plans
- `/demo` - Demo signup page
- `/onboarding` - New tenant onboarding with Transbank payment
- `/auth/login` - Login page
- `/auth/register` - Registration page
- `/auth/signout` - Custom signout page (NEW)
- `/privacy` - Privacy policy (NEW)
- `/terms` - Terms of service (NEW)
- `/subscription` - Subscription management
- `/subscriptions/plans` - Available plans
- `/subscriptions/payment/success` - Payment success
- `/subscriptions/payment/error` - Payment error

Tenant Admin Routes (`/admin/*`)

Protected by middleware - requires authentication

- `/admin/dashboard` - Main dashboard with metrics
- `/admin/pos` - Point of Sale interface
- `/admin/inventory` - Inventory management
- `/admin/inventory/add-from-pool` - Add from master products
- `/admin/inventory/movements` - View stock movements
- `/admin/cash-session` - Cash register management
- `/admin/sales` - Sales history and management
- `/admin/reports` - Business intelligence reports
- `/admin/reports/sales` - Sales reports
- `/admin/reports/products` - Product reports
- `/admin/reports/customers` - Customer reports
- `/admin/settings` - Tenant configuration (NEW)

SaaS Admin Routes (`/admin-saas/*`)

Protected - PROVEEDOR role only

- `/admin-saas` - Admin dashboard
- `/admin-saas/tenants` - Customer management ("Gestor de clientes")
- `/admin-saas/tenants/[id]` - Individual tenant details

- `/admin-saas/subscriptions` - Subscription management
- `/admin-saas/subscriptions/[id]` - Individual subscription
- `/admin-saas/plans` - Subscription plan CRUD (NEW)
- `/admin-saas/revenue` - Revenue analytics
- `/admin-saas/master-products` - Master product catalog CRUD
- `/admin-saas/stats` - Platform statistics

Provider Routes (`/provider/*`)

- `/provider/products` - Product management

Deprecated Routes

- `/saas-admin.deprecated/*` - Old admin routes (moved to `/admin-saas`)

API Routes (app/api/)

Authentication

- `POST /api/auth/[...nextauth]` - NextAuth endpoints
- `POST /api/auth/register` - User registration

Public APIs

- `GET /api/public/plans` - Public plan listing (NEW)
- `POST /api/demo` - Demo account creation
- `POST /api/onboarding` - Tenant onboarding

Tenant Admin APIs

• Cash Sessions:

- `GET/POST /api/cash-sessions`
- `POST /api/cash-sessions/[id]/close`
- `GET /api/cash-sessions/active`

• Inventory:

- `GET/POST /api/inventory`
- `PUT/DELETE /api/inventory/[id]`
- `GET /api/inventory/available-products`
- `GET/POST /api/inventory/movements`

• Products (Legacy):

- `GET/POST /api/products`
- `GET/PUT/DELETE /api/products/[id]`

• Sales:

- `GET/POST /api/sales`
- `GET /api/sales/[id]`
- `GET /api/sales/stats`

• Reports:

- `GET /api/reports/sales`
- `GET /api/reports/products`

- GET /api/reports/customers
- GET /api/reports/export
- **Settings (NEW):**
 - GET/PUT /api/settings/company
 - GET/PUT /api/settings/notifications
 - GET/PUT /api/settings/pos
 - GET/PUT /api/settings/subscription
 - POST /api/settings/subscription/cancel
 - POST /api/settings/subscription/upgrade
 - GET/POST /api/settings/users

SaaS Admin APIs

- **Tenants:**
 - GET/POST /api/admin-saas/tenants
 - GET/PUT/DELETE /api/admin-saas/tenants/[id]
 - POST /api/admin-saas/tenants/[id]/activate
 - POST /api/admin-saas/tenants/[id]/suspend
 - POST /api/admin-saas/tenants/[id]/change-plan
 - GET /api/admin-saas/tenants/[id]/products
 - GET /api/admin-saas/tenants/[id]/users
- **Master Products:**
 - GET/POST /api/admin-saas/master-products
 - GET/PUT/DELETE /api/admin-saas/master-products/[id]
- **Metrics:**
 - GET /api/admin-saas/metrics
 - GET /api/admin-saas/stats
- **Plans (NEW):**
 - GET/POST /api/saas/plans
 - GET/PUT/DELETE /api/saas/plans/[id]
- **Subscriptions:**
 - GET/POST /api/saas/subscriptions
 - GET /api/saas/subscriptions/recent
 - GET /api/saas/subscriptions/renewals
 - GET /api/saas/revenue
- **Subscription Plans:**
 - GET/POST/PUT/DELETE /api/subscription-plans
 - GET/PUT/DELETE /api/subscription-plans/[id]

Payment APIs (Transbank Integration)

- `POST /api/payments/initiate` - Start payment
- `GET /api/payments/confirm` - Confirm payment
- `GET /api/payments/history` - Payment history
- `POST /api/subscriptions/payment/init`
- `POST /api/subscriptions/payment/callback`

System APIs

- `POST /api/init-db` - Database initialization
- `POST /api/cron/subscription-tasks` - Automated subscription tasks



Component Architecture

UI Components (`components/ui/`)

Based on **shadcn/ui** library:

- `button`, `card`, `dialog`, `dropdown-menu`
- `input`, `label`, `select`, `checkbox`
- `table`, `tabs`, `toast`, `toaster`
- `avatar`, `badge`, `separator`, `skeleton`
- `accordion`, `alert-dialog`, `scroll-area`
- Plus ~30 more UI primitives

Business Components

Admin Components (`components/admin/`)

- **AdminNavBar.tsx** - Main navigation with mobile menu
- Menu items: Dashboard, POS, Inventory, Cash Session, Sales, Reports
- User dropdown with settings
- Role-based menu filtering
- **BackButton.tsx** - Navigation helper

Landing Components (`components/landing/`)

- **PricingPlans.tsx** - Public pricing display (NEW)
- Fetches plans from `/api/public/plans`
- Shows price, features, trial days
- Call-to-action buttons

SaaS Admin Components (`components/saas-admin/`)

- **MetricsCards.tsx** - KPI cards for dashboard
- **PlanDistributionChart.tsx** - Visual plan distribution
- **PlanManagement.tsx** - Plan CRUD interface
- **RecentSubscriptions.tsx** - Latest subscriptions
- **RevenueChart.tsx** - Revenue visualization
- **UpcomingRenewals.tsx** - Renewal tracking

Other Components

- **SubscriptionStatusBanner.tsx** - Status alerts

- **providers.tsx** - NextAuth session provider
- **theme-provider.tsx** - Dark/light theme

Authentication & Authorization

NextAuth Configuration (`lib/auth.ts`)

- **Provider:** Credentials-based authentication
- **Adapter:** PrismaAdapter for database sessions
- **Session Strategy:** JWT tokens
- **Custom Session Data:**
 - `id` , `firstName` , `lastName`
 - `role` , `tenantId`

Middleware (`middleware.ts`)

- **Protected Routes:**
 - `/admin/*` - Tenant admin routes
 - `/caja/*` , `/inventario/*` , `/soporte/*`
 - `/saas/*` , `/saas-admin/*` , `/admin-saas/*`
- **Redirect:** Unauthenticated users → `/auth/login`

Role-Based Access

1. **PROVEEDOR** - Platform admin, full access
2. **ADMIN** - Tenant admin, business management
3. **CAJA** - Cashier, POS operations
4. **INVENTARIO** - Inventory manager
5. **SOPORTE** - Support staff

Authorization Checks

- **Frontend:** Layout components check `session.user.role`
- **API:** Routes validate user role and `tenantId`
- **Example:** Admin-SaaS routes require `role === 'PROVEEDOR'`

Deployment Configuration

Cloud Run Services

1. **crtlpyme** - <https://crtlpyme-399888129827.us-central1.run.app>
2. **crtlpyme-app** - <https://crtlpyme-app-399888129827.us-central1.run.app>

Dockerfile Analysis

- **Base:** `node:18-alpine`
- **Multi-stage build:** `deps` → `builder` → `runner`
- **Build Process:**
 1. Install dependencies with `--legacy-peer-deps`
 2. Generate Prisma Client

3. Build Next.js app
4. Copy standalone output

- **Runtime:**

- User: `nextjs` (non-root)
- Port: 3000
- Command: `node server.js`

- **Critical Files Copied:**

- `.next/standalone`
- `.next/static`
- `node_modules/.prisma`
- `prisma/` directory

GitHub Actions Workflow (`.github/workflows/deploy.yml`)

- **Trigger:** Push to `main / master` or manual dispatch

- **Steps:**

1. Checkout code
2. Authenticate with GCP using `GCP_SA_KEY` secret
3. Configure Docker for Artifact Registry
4. Build Docker image with tags:
 - `:${{ github.sha }}`
 - `:latest`
5. Push to `us-central1-docker.pkg.dev`
6. Deploy to Cloud Run with:
 - Memory: 2Gi, CPU: 2
 - Timeout: 300s
 - Autoscaling: 0-10 instances
 - Port: 3000
7. Set secrets from Google Secret Manager:
 - `DATABASE_URL` , `NEXTAUTH_SECRET` , `NEXTAUTH_URL`
 - `SENDGRID_API_KEY` , `SENDGRID_FROM_EMAIL`
 - `TRANSBANK_API_KEY` , `TRANSBANK_COMMERCE_CODE` , `TRANSBANK_ENVIRONMENT`
8. Set environment variables
9. Verify deployment (HTTP 200/301/302)

Database Configuration

- **Type:** Cloud SQL PostgreSQL
 - **Database:** `ctrlpyme`
 - **User:** `postgres`
 - **Password:** `CRTLPyne2025!`
 - **Connection:** via `DATABASE_URL` secret
-



Environment Variables Required

Database

```
DATABASE_URL="postgresql://postgres:CTRLPyme2025!@<host>/ctrlpyme"
```

NextAuth

```
NEXTAUTH_SECRET="<secret-key>"
NEXTAUTH_URL="https://ctrlpyme-app-399888129827.us-central1.run.app"
```

Email (SendGrid)

```
SENDGRID_API_KEY="<api-key>"
SENDGRID_FROM_EMAIL="noreply@ctrlpyme.cl"
SENDGRID_FROM_NAME="CTRLPyme"
```

Transbank Payment

```
TRANSBANK_COMMERCE_CODE="<commerce-code>"
TRANSBANK_API_KEY="<api-key>"
TRANSBANK_ENVIRONMENT="production" # or "integration"
```

Cron Security

```
CRON_SECRET="<random-secret>"
```

System

```
NODE_ENV="production"
NEXT_PUBLIC_APP_NAME="CTRLPyme"
GOOGLE_CLOUD_PROJECT_ID="ctrlpyme-477300"
```



Recent Changes (Git History)

Latest Commits (Nov 8, 2025)

1. 063fa53 - UI text improvements (08:50 AM)

- Changed “Gestión de Tenants” → “Gestor de clientes”
- Created custom signout page with Spanish translations
- Updated copyright year 2024 → 2025
- **Files:**
 - `app/admin-saas/tenants/page.tsx`
 - `app/auth/signout/page.tsx` (NEW)
 - `app/page.tsx` , `app/privacy/page.tsx` , `app/terms/page.tsx`
 - `lib/auth.ts`

2. bf7977e - Navigation fixes and legal pages (08:35 AM)

- Fixed routes: `/auth/signup` → `/auth/register`
- Fixed PROVEEDOR redirect: `/saas-admin` → `/admin-saas`
- Added legal pages: `/terms` and `/privacy`
- Consolidated admin panel (deprecated old `/saas-admin`)
- **Mobile responsiveness improvements:**
 - Responsive mobile menu
 - Hidden sidebar on mobile
 - Mobile header with overlay menu
- **Documentation added:**
 - `INTERFACE_AUDIT_REPORT.md`
 - `MOBILE_RESPONSIVENESS_ANALYSIS.md`
 - `PHASE6_IMPLEMENTATION_SUMMARY.md`
 - `PLANS_IMPLEMENTATION_SUMMARY.md`

3. 0c7386b - Plans on landing page (08:01 AM)

- Added pricing section to landing page
- Created `PricingPlans` component
- Added `/api/public/plans` endpoint
- Full CRUD for subscription plans in admin-saas
- **Features:**
 - Visible/hidden plans toggle
 - Active/inactive status
 - Plan deletion with safety checks

4. 307cc44 - Tenant admin settings (07:xx AM)

- Implemented tenant configuration panel
- Added settings pages and APIs:
 - Company settings
 - Notification preferences
 - POS configuration
 - Subscription management
 - User management

5. Previous Major Features

- **c88100d** - Demo and onboarding with Transbank
 - **5e3dfcc** - Master product CRUD for providers
 - **803f258** - Inventory movement tracking system
 - **d866152** - TenantInventory for sales (not Product)
 - **8db934d** - Migration from GCR to Artifact Registry
 - **856d6ef** - CI/CD workflow for Cloud Run
-

Complete Feature List

Core Features (Implemented)

1. Multi-Tenant System

- [x] Tenant registration and onboarding
- [x] Tenant isolation (all queries filtered by tenantId)
- [x] Subscription plan management
- [x] Account status tracking (ACTIVE, TRIAL, SUSPENDED, BLOCKED)

2. Authentication & Authorization

- [x] Email/password login
- [x] Role-based access control (5 roles)
- [x] JWT session management
- [x] Protected routes middleware
- [x] Custom signout page

3. Product Management

- [x] Master Product Catalog (shared pool)
- [x] Tenant-specific inventory (TenantInventory)
- [x] Barcode support (EAN-13)
- [x] Stock tracking
- [x] Stock adjustment records
- [x] Inventory movement tracking
- [x] Add products from master pool
- [x] Low stock alerts (minStock threshold)

4. Point of Sale (POS)

- [x] Complete POS interface
- [x] Barcode scanning support
- [x] Multiple payment methods (CASH, DEBIT, CREDIT, TRANSFER)
- [x] Real-time stock updates
- [x] Sale receipt generation

5. Cash Management

- [x] Cash session opening/closing
- [x] Initial amount tracking
- [x] Expected vs actual amount
- [x] Discrepancy reporting
- [x] Session history

6. Sales Management

- [x] Sale recording with line items
- [x] Sale number generation (per tenant)
- [x] Sale status tracking
- [x] Sale history and search
- [x] Profit margin calculation (unitCost vs unitPrice)

7. Reporting & Analytics

- [x] Sales reports
- [x] Product performance reports
- [x] Customer reports
- [x] Export functionality
- [x] Dashboard with KPIs

8. SaaS Administration

- [x] Platform admin dashboard
- [x] Tenant management (CRUD)
- [x] Subscription management
- [x] Plan management (CRUD)
- [x] Master product management
- [x] Revenue analytics
- [x] Platform statistics
- [x] Tenant action logging
- [x] Tenant suspension/activation

9. Subscription & Billing

- [x] Subscription plans with features
- [x] Trial period support
- [x] Billing cycle management (MONTHLY, QUARTERLY, ANNUAL)
- [x] Auto-renewal logic
- [x] Subscription cancellation
- [x] Plan upgrades/downgrades
- [x] Payment history

10. Payment Integration (Transbank)

- [x] Payment initiation
- [x] Payment confirmation
- [x] Webhook handling
- [x] Refund processing
- [x] Payment history
- [x] Card information storage (last 4 digits)

11. Notifications & Email

- [x] Email template system
- [x] Email queue with retry logic
- [x] Notification preferences
- [x] Notification history
- [x] SendGrid integration

12. Audit & Security

- [x] Complete audit logging
- [x] Action tracking (CREATE, UPDATE, DELETE)
- [x] Old/new value comparison
- [x] User action attribution

13. User Management

- [x] User creation
- [x] Role assignment
- [x] User activation/deactivation
- [x] User profile management

14. Business Intelligence

- [x] Fixed expense tracking
- [x] Break-even analysis
- [x] Revenue calculations
- [x] Platform metrics
- [x] Subscription metrics
- [x] Dashboard snapshots (caching)

15. Mobile Responsiveness

- [x] Mobile-friendly navigation
- [x] Responsive layouts
- [x] Mobile menu overlay
- [x] Touch-friendly controls

16. Legal & Compliance

- [x] Privacy policy page
- [x] Terms of service page
- [x] Chilean market compliance



Known Issues & Concerns

Critical Issues

1. Login Not Working

Symptom: User reports login doesn't work with any account

Potential Causes:

- Database connection issue (DATABASE_URL misconfigured)
- NextAuth secret mismatch (NEXTAUTH_SECRET not matching)
- NEXTAUTH_URL incorrect (should be deployment URL)
- No users in production database
- Password hashing mismatch (bcryptjs)
- Session secret not properly set

Files to Check:

- `lib/auth.ts` - Auth configuration
- `.env` secrets in Google Secret Manager
- `app/api/auth/[...nextauth]/route.ts`

Recommended Actions:

1. Verify DATABASE_URL connects to correct Cloud SQL instance
2. Check NEXTAUTH_SECRET is set correctly
3. Ensure NEXTAUTH_URL matches deployed URL

4. Run seed script to create test users
5. Check Cloud Run logs for auth errors

2. Missing Features in Deployment

Symptom: Recent features (menu, components) missing from deployed version

Potential Causes:

- **Build cache issue** - Docker using old layer
- **Deployment failed silently** - CI/CD didn't error
- **Wrong service deployed** - `crtlpyme` vs `crtlpyme-app`
- **Git not pushed** - Latest commits not in repo
- **Build artifact incomplete** - Standalone build missing files

Files to Check:

- `.github/workflows/deploy.yml` - CI/CD configuration
- `Dockerfile` - Build process
- `next.config.js` - Output configuration

Recommended Actions:

1. Verify latest commit (063fa53) is deployed
2. Check which service is actually running
3. Review Cloud Run logs for build errors
4. Force rebuild without cache
5. Verify `.next/standalone` includes all files

3. Database Schema Mismatch

Potential Issue: Production database may not have latest migrations

Evidence:

- Recent changes to TenantInventory usage
- New tables added (settings, notifications)
- Schema changes for admin-saas features

Recommended Actions:

1. Check `prisma/migrations` directory
2. Run `prisma migrate status` against production DB
3. Apply pending migrations
4. Verify all tables exist

Medium Priority Issues

4. Service URL Confusion

- Two Cloud Run services exist: `crtlpyme` and `crtlpyme-app`
- Unclear which one is the primary service
- Different URLs may show different versions

5. Session Management

- Sessions stored in database (PrismaAdapter)
- Could cause issues if DB connection drops
- No session cleanup mechanism visible

6. Legacy Product Table

- `Product` table marked as DEPRECATED
- Code may still reference it in places
- Migration to `TenantInventory` may be incomplete

7. Transbank Integration

- Environment variable `TRANSBANK_ENVIRONMENT` determines test vs prod
- Unclear if production keys are configured
- Webhook security not verified

Low Priority Issues

8. TypeScript & ESLint Disabled

```
// next.config.js
typescript: { ignoreBuildErrors: true },
eslint: { ignoreDuringBuilds: true },
```

Risk: Type errors and linting issues may exist in production

9. No Database Connection Pooling

- Prisma client initialized directly
- Could cause connection exhaustion under load

10. Hardcoded Business Logic

- Some tenant limits hardcoded
- Magic numbers in code
- Should be configurable



Deployment Troubleshooting Guide

Issue: Login Not Working

Step 1: Verify Database Connection

```
# From Cloud Run console, check logs for:
"Prisma Client could not connect to database"
"Invalid DATABASE_URL"
```

Step 2: Check Secrets

```
# List secrets in Google Secret Manager
gcloud secrets list --project crtlpyme-477300

# View secret values (requires permission)
gcloud secrets versions access latest --secret="DATABASE_URL"
gcloud secrets versions access latest --secret="NEXTAUTH_SECRET"
gcloud secrets versions access latest --secret="NEXTAUTH_URL"
```


Step 3: Test Database Query

```
# SSH into Cloud SQL and check users table
gcloud sql connect <instance-name> --user=postgres --database=ctrlpyme
SELECT email, role, "isActive" FROM users LIMIT 5;
```

Step 4: Create Test User

Run seed script or manually insert:

```
-- Test user credentials
-- Email: admin@test.com
-- Password: admin123
-- Check if bcrypt hash matches
```

Issue: Missing Features in Deployment

Step 1: Verify Git Commit Deployed

Check GitHub Actions run for commit SHA:

```
Current SHA: 063fa53
Deployed SHA: <check Cloud Run image tag>
```

Step 2: Check Cloud Run Revision

```
gcloud run revisions list --service=ctrlpyme-app --region=us-central1
```

Step 3: Force New Deployment

```
# Trigger manual workflow run
# Or push a dummy commit to trigger CI/CD
```

Step 4: Review Build Logs

```
# Check GitHub Actions logs
# Look for:
# - npm install errors
# - prisma generate errors
# - next build errors
```

Issue: Database Schema Out of Sync

Step 1: Check Migration Status

```
# From local machine with DB access
npx prisma migrate status --schema=./prisma/schema.prisma
```

Step 2: Apply Migrations

```
# CAUTION: Test on staging first
npx prisma migrate deploy
```

Step 3: Generate Prisma Client

```
# Ensure client matches schema
npx prisma generate
```



Recommended Immediate Actions

For User's Urgent Presentation

1. Verify Current Deployment Status

- Check which service is running (crtlpyme vs crtlpyme-app)
- Confirm latest commit is deployed
- Review Cloud Run logs for errors

2. Fix Login Issue

- Verify database connectivity
- Check NEXTAUTH_URL matches deployment
- Create test user account if none exist
- Test login with known credentials

3. Verify Features Present

- Access landing page - should show pricing plans
- Login as PROVEEDOR - should see admin-saas menu
- Check for: Plans, Tenants, Subscriptions, Master Products
- Login as ADMIN - should see tenant admin features

4. Create Demo Data (if missing)

```
bash
# Run comprehensive seed script
npm run seed:complete
```

5. Document Known Limitations

- Prepare list of "known issues" for presentation
- Have screenshots ready as backup
- Prepare explanation of architecture



Additional Documentation Files in Repo

The repository contains extensive documentation:

Implementation Summaries

- `FASE1_FINAL_SUMMARY.md` - Phase 1 completion
- `PHASE5_IMPLEMENTATION_SUMMARY.md` - Phase 5 details
- `PHASE6_IMPLEMENTATION_SUMMARY.md` - Phase 6 (latest)

- `PLANS_IMPLEMENTATION_SUMMARY.md` - Plan management

Deployment Guides

- `DEPLOYMENT_GUIDE.md` - General deployment
- `CICD_SETUP_GUIDE.md` - CI/CD configuration
- `GCP_MIGRATION_GUIDE.md` - GCP setup
- `INSTRUCCIONES_DEPLOYMENT_PRODUCCION.md` - Production deployment

Technical Documentation

- `README-CICD.md` - CI/CD details
- `README_ADMIN_SAAS.md` - Admin features
- `README_SEED.md` - Seed data guide
- `SEED_DOCUMENTATION.md` - Seed scripts

Analysis Reports

- `INTERFACE_AUDIT_REPORT.md` - UI audit
- `MOBILE_RESPONSIVENESS_ANALYSIS.md` - Mobile analysis
- `DATABASE_VERIFICATION_SUMMARY.md` - DB verification
- `CLEANUP_SUMMARY.md` - Code cleanup

Feature Documentation

- `TRANSBANK_IMPLEMENTATION_GUIDE.md` - Payment integration
- `INVENTORY_MOVEMENTS_IMPLEMENTATION.md` - Inventory tracking
- `Modulo_Admin_SaaS_CRTLPyme.md` - Admin module
- `Preparacion_POS_CRTLPyme.md` - POS preparation

Summary of What Should Be Working

Based on the code analysis, the deployed application **should have**:

Public Features

- Landing page with hero section
- Pricing plans display (3 tiers)
- Demo signup button
- Login page
- Registration page
- Privacy policy
- Terms of service

Tenant Admin Features (ADMIN role)

- Dashboard with sales metrics
- Point of Sale interface
- Inventory management
- Add products from master catalog
- View/edit stock levels
- Track inventory movements

- Cash session management
- Sales history
- Reports (sales, products, customers)
- Settings panel
- Company info
- Notifications
- POS configuration
- Subscription management

✓ **SaaS Admin Features (PROVEEDOR role)**

- Admin dashboard
- Customer management (“Gestor de clientes”)
- Subscription management
- Plan CRUD (create/edit/delete plans)
- Master product catalog
- Revenue analytics
- Platform statistics

✓ **Payment Features**

- Transbank integration for subscriptions
- Payment history
- Refund processing

✓ **System Features**

- Multi-tenant data isolation
- Role-based access control
- Audit logging
- Email notifications
- Mobile responsive design

Technical Specifications

Frontend

- **Framework:** Next.js 15.0.3
- **UI Library:** shadcn/ui (Radix UI + Tailwind)
- **Language:** TypeScript 5
- **Styling:** Tailwind CSS 3.4.1
- **Icons:** Lucide React
- **Charts:** Recharts 3.3.0

Backend

- **API:** Next.js API Routes
- **ORM:** Prisma 6.0.1
- **Database:** PostgreSQL 15
- **Authentication:** NextAuth.js 4.24.10

Deployment

- **Platform:** Google Cloud Run
 - **Container:** Docker (node:18-alpine)
 - **CI/CD:** GitHub Actions
 - **Image Registry:** Artifact Registry
 - **Database:** Cloud SQL PostgreSQL
-



Code Statistics

- **Total Files:** 394
 - **TypeScript/TSX Files:** 196
 - **Pages/Routes:** 40+ frontend routes
 - **API Endpoints:** 60+ API routes
 - **Database Tables:** 30 tables
 - **Enums:** 20+ Prisma enums
 - **UI Components:** 40+ shadcn components
 - **Business Components:** 15+ custom components
-



Project Context

Purpose: Thesis project (Capstone 707V)

Team: Hernán Cabezas & Grisel Sanchez

Professor: Fernando González

Target: Chilean SMBs (PYMEs)

Market: Chilean retail (abarrotes, kioscos)

Year: 2025



Important Links

- **GitHub:** <https://github.com/kbzas090/CRTLPyme>
 - **Production (App):** <https://crtlpyme-app-399888129827.us-central1.run.app>
 - **Production (Alt):** <https://crtlpyme-399888129827.us-central1.run.app>
 - **GCP Project:** crtlpyme-477300
 - **Region:** us-central1
-



Contact

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End of Analysis

Generated: November 8, 2025

By: AI Code Analysis System